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M1 MACRO

Group A

Session II

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Iterated Expectations

$$\underbrace{E_t \left[\underbrace{E_{t+1} [X_{t+2}]} \right]} = \underbrace{E_t [X_{t+2}]}$$

Take an example

Dice

$\frac{1}{2} \rightarrow$ equiprobable $p_i = \frac{1}{6}$ (Dice A)

$\frac{1}{2} \rightarrow$ not equiprobable $p(1) = \frac{1}{2}$
(Dice B) $p(2,3,4,5,6) = \frac{1}{10}$

$t=0$

$t=1$

$t=2$

choose the dice

roll the dice

$E_0(X_2)$

random with
prob $\frac{1}{2}$

X_2

$E_1(X_2)$

$$E_1(X_2) = ?$$

$$\text{if Dice A : } E_1(X_2 / \text{Dice A}) = 3.5$$

$$\text{if Dice B } E_1(X_2 / \text{Dice B}) = \frac{1}{2} + \frac{1}{10}(2+3+4+5+6) = 2.5$$

$$E_0[X_2] = \frac{1}{2} \underbrace{\left[\frac{1}{6} (1+2+3+4+5+6) \right]}_{\text{Dice A}} + \frac{1}{2} \underbrace{\left[\frac{1}{2} + \frac{1}{10} (2+3+4+5+6) \right]}_{\text{Dice B}}$$

$$E_0(X_2) = \frac{1}{2} \cdot 3.5 + \frac{1}{2} \cdot 2.5 = \underline{\underline{3}}$$

$$E_0(E_1(X_2))?$$

$$= \frac{1}{2} \times E_1(X_2 / \text{Dice } A) + \frac{1}{2} \times E_1(X_2 / \text{Dice } B)$$

$$= \frac{1}{2} \cdot 3.5 + \frac{1}{2} \cdot 2.5 = \underline{\underline{3}}$$

$$E_0(E_1(X_2)) = E_0(X_2)$$

Solving
$$M_t = \frac{1}{1+r} E \sum_{i=0}^{\infty} \frac{y_{t+i}}{(1+r)^i}$$

Assume
$$\begin{cases} y_t = \bar{y} + \tilde{y}_t \\ \tilde{y}_t = \rho \tilde{y}_{t-1} + \varepsilon_t \end{cases} \quad \left(E(\varepsilon_t) = 0 \right)$$

Simple case
$$\boxed{\rho = 0} \Rightarrow \tilde{y}_t = \varepsilon_t$$

$$E_t(\tilde{y}_{t+i}) = 0 \quad \forall i > 0$$

Therefore,

$$\tilde{y}_{t+i} = \varepsilon_{t+i}$$

$$H_t = \frac{1}{1+\alpha} \sum_{i=0}^{\infty} \frac{\bar{y} + E(\varepsilon_{t+i})}{(1+\alpha)^i}$$

$$= \frac{1}{1+\alpha} \left[\sum_{i=0}^{\infty} \frac{\bar{y}}{(1+\alpha)^i} + \left[\varepsilon_t + \sum_{i=1}^{\infty} E(\varepsilon_{t+i}) \right] \right]$$

$$H_t = \frac{\bar{y}}{\alpha} + \frac{\varepsilon_t}{1+\alpha}, \text{ so that}$$

$$C_t = \alpha A_t \bar{y} + \frac{\alpha}{1+\alpha} \varepsilon_t$$

$$C_t = n A_t + \underbrace{\overline{y}}_{\text{permanent income}} + \frac{n}{1+r} \underbrace{\varepsilon_t}_{\text{temporary income}}$$

marginal propensity to consume

$$\frac{\partial C}{\partial \overline{y}} = 1, \quad \frac{\partial C}{\partial \varepsilon_t} = \frac{n}{1+r} \ll 1$$

No simple case $1 > \rho > 0$

$$W_t = \frac{1}{1+\alpha} \sum_{i=0}^{\infty} \left(\bar{y} + \tilde{y}_t \right) * \frac{1}{(1+\alpha)^i}$$

$$= \frac{\bar{y}}{n} + \frac{1}{1+\alpha} \sum_{t=0}^{\infty} \frac{E_t \tilde{y}_{t+i}}{(1+\alpha)^i}$$

$$\tilde{y}_{t+1} = \rho \tilde{y}_t + \varepsilon_{t+1}$$

$$E_t \tilde{y}_{t+1} = \rho \tilde{y}_t$$

$$E_t \tilde{y}_{t+2} = E_t \left[\rho^2 \tilde{y}_t + \varepsilon_{t+2} + \rho \varepsilon_{t+1} \right] = \rho^2 \tilde{y}_t$$

$$\Rightarrow \boxed{E_t [\tilde{y}_{t+i}] = \rho^i \tilde{y}_t}$$

Therefore $H_t = \frac{\bar{y}}{n} + \frac{1}{1+\alpha} \sum_{i=0}^{\infty} \frac{\rho^i}{(1+\alpha)^i} \tilde{y}_t$

$$= \frac{\bar{y}}{n} + \frac{\tilde{y}_t}{1+\alpha} \times \sum_{i=0}^{\infty} \left(\frac{\rho}{1+\alpha} \right)^i$$

$$\underbrace{\frac{1}{1 - \frac{\rho}{1+\alpha}}} = \frac{1+\alpha}{1+\alpha-\rho}$$

$$H_t = \frac{\bar{y}}{n} + \frac{\tilde{y}_t}{1+\alpha-\rho}$$

$$M_t = \frac{\overline{y}}{n} + \frac{\tilde{y}_t}{1+n-p}$$

permanent
component
of income

temporary component
of income.

$$C_t = nA_t + \overline{y} + \frac{n}{1+n-p} \tilde{y}_t$$

• if $p=0$ $\frac{\partial C_t}{\partial \tilde{y}_t} = \frac{n}{1+n}$

• if $p=1$ $\frac{\partial C_t}{\partial \tilde{y}_t} = 1$

$$\tilde{y}_{t+1} = \rho \tilde{y}_t + \varepsilon_{t+1}$$

ρ = persistence of move

$\frac{\partial C_F}{\partial \tilde{y}_t}$ is increasing in move persistence